Statement of Purpose: Amazon Machine Learning Summer School

I�ve always hated treating machine learning models like magic black boxes. As a second-year AI & Data Science student at IIITDM Kurnool, my biggest priority has been figuring out how things actually work under the hood. That�s what pushed me to build the original Transformer architecture entirely from scratch using pure PyTorch. I wanted to genuinely understand multi-head attention and causal masking, rather than just calling a HuggingFace API and hoping for the best.

That curiosity quickly spilled over into building full-scale projects. Recently, I built *CourtSense-AI*, a computer vision pipeline that takes 2D tennis match videos and turns them into 3D interactive replays. I used YOLOv8-Pose and SegFormer, but the real challenge was optimizing it with TensorRT to hit 30+ FPS on edge hardware. I also wanted to get my hands dirty with multimodal AI, so I built my own version of the PaliGemma vision-language model from the ground up�hooking up a SigLIP encoder to a Gemma-2B decoder and using 8-bit quantization to cut VRAM usage in half.

Right now, I�m working as a Research Intern at SVNIT Surat, focusing on the security side of AI. I recently discovered that if you execute a surgical adversarial attack on just the Stage-4 attention blocks of a SegFormer model, its accuracy absolutely crashes�from 97% down to 2% on the Market-1501 dataset. We call it a "Nuclear Attack," and the research paper is currently under review. But I don't just keep these things to myself; as the ML Coordinator for my campus GDG, I recently led a 12-week bootcamp where I taught over 200 students how to go from learning basic gradient descent to shipping real-world ML apps.

Despite everything I�ve built, there is still a massive gap in my knowledge that I�m hoping the Amazon MLSS can help me fill. I know how to train and deploy complex architectures on local GPUs and edge devices, but I have almost no experience scaling these models to a massive, distributed enterprise level. I want to learn Amazon�s actual industry standards for things like distributed training, automating robust MLOps pipelines, and handling incredibly large, noisy datasets in the cloud.

I believe I�d be a great fit for this program because I have a proven track record of diving into the deep end and learning fast. I�m the kind of developer who tears architectures apart to find their bottlenecks rather than just using them blindly. I would absolutely love the opportunity to absorb Amazon�s expertise in scalable AI, use it to push my research further, and eventually share those enterprise-grade insights with the students I mentor.